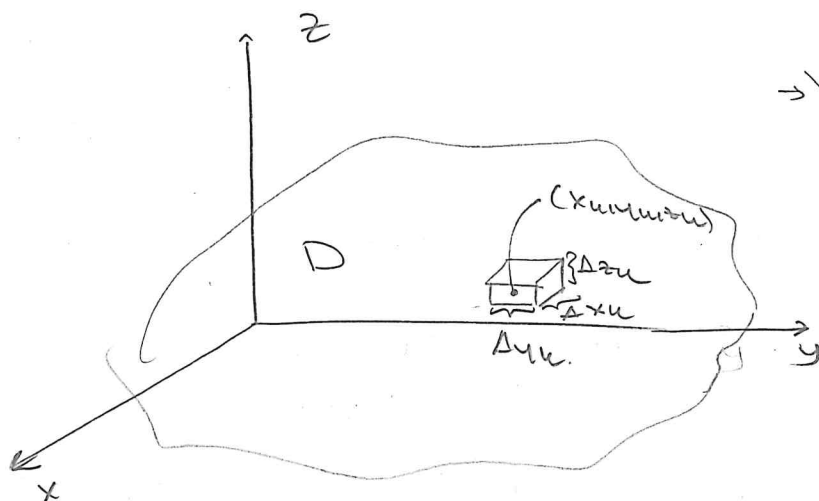


CALCULUS ④

WEEK XIV

TRIPLE INTEGRALS

→ $F(x, y, z)$ is a function defined on a closed, bounded region D in space. We partition a rectangular box like region containing D into rectangular cells by planes parallel to the coordinate axes.



→ We form the sum

$$S_n = \sum_{k=1}^n F(x_k, y_k, z_k) \Delta V_k$$

As the norm of partition approach to zero, if a single limiting value is attained no matter how the partition and points are chosen, we say that F is integrable over D .

$$\lim_{n \rightarrow \infty} S_n = \iiint_D F(x, y, z) dV \quad \text{or} \quad \lim_{\|P\| \rightarrow 0} S_n = \iiint_D F(x, y, z) dx dy dz$$

the norm of the partition $\|P\|$ = the largest value among $\Delta x_k, \Delta y_k, \Delta z_k$

Def The volume of a closed, bounded region D in space is

$$V = \iiint_D dV$$

Ex Evaluate the following integral

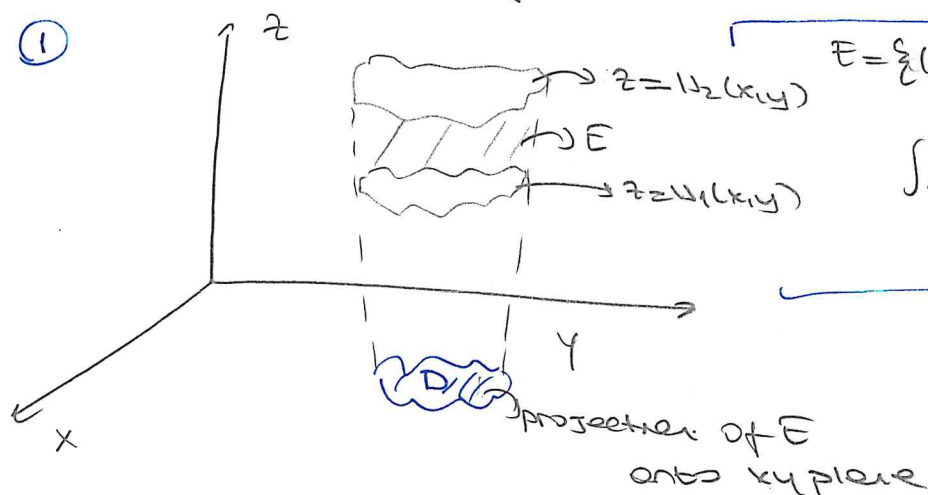
$$\iiint_B 8xyz \, dV = ?$$

$$B = \underbrace{[2, 3]}_x \times \underbrace{[1, 2]}_y \times \underbrace{[0, 1]}_z$$

$$\begin{aligned} \iiint_B 8xyz \, dV &= \int_1^2 \int_2^3 \int_0^1 8xyz \, dz \, dx \, dy = \int_1^2 \int_2^3 (4xy z^2) \Big|_0^1 \, dx \, dy \\ &= \int_1^2 \int_2^3 4xy \, dx \, dy = \int_1^2 (2x^2 y) \Big|_2^3 \, dy = \int_1^2 10y \, dy = 5y^2 \Big|_1^2 = 15 \end{aligned}$$

We have three different possibilities for a good region

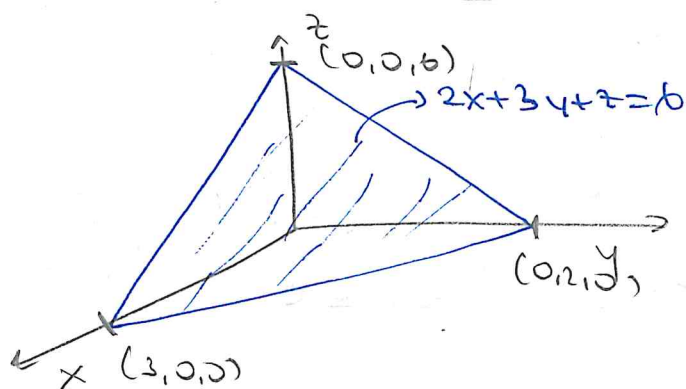
①



$$E = \{(x, y, z) : (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y)\}$$

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz \right]$$

Ex 7 Evaluate $\iiint_E 2x dV$ where E is the region under the plane $2x + 3y + z = 6$ that lies in the first octant ($x \geq 0, y \geq 0, z \geq 0$).



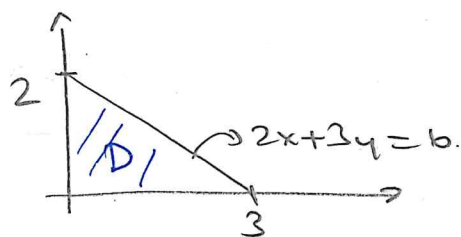
$$2x + 3y + z = 6 \Rightarrow x = y = 0 \quad z = 6$$

$$x = z = 0 \quad y = 2$$

$$y = z = 0 \quad x = 3$$

$$0 \leq z \leq 6 - 2x - 3y$$

We now determine the region D in the xy -plane. It is a triangle region in the xy -plane with vertices at $(0, 0)$ and $(3, 0)$.



$$\iint_D \left[\int_{z=0}^{z=6-2x-3y} 2x dz \right] dA =$$

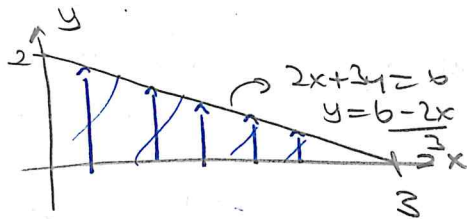
$$\iint_D 2x \cdot z \Big|_{z=0}^{z=6-2x-3y} dA =$$

$$\iint_D 2x (6 - 2x - 3y) dA = \iint_D (12x - 4x^2 - 6xy) dA$$

②

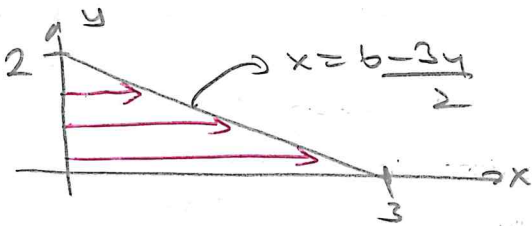
Consider double integral

①



$$\iiint_D \left[\int_{z=0}^{z=b-2x-3y} 2x \, dz \right] dA = \int_0^3 \int_0^{b-\frac{2x}{3}} \int_{z=0}^{z=b-2x-3y} 2x \, dz \, dy \, dx$$

②



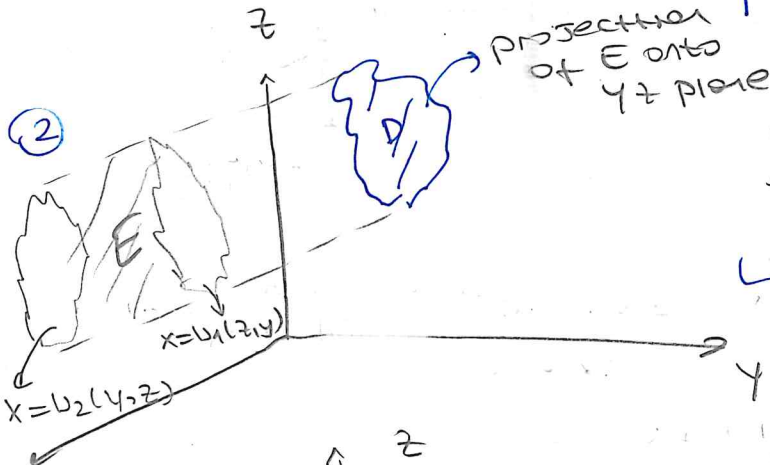
$$\iiint_D \left[\int_{z=0}^{z=b-2x-3y} 2x \, dz \right] dx \, dy =$$

$$\int_0^3 \int_0^{b-\frac{2x}{3}} \left[\int_{z=0}^{z=b-2x-3y} 2x \, dz \right] dy \, dx = \int_0^3 \int_b^{\frac{b-2x}{3}} 2x (b-2x-3y) \, dy \, dx$$

$$= \int_0^3 \left[12xy - 4x^2y - 3xy^2 \right] \Big|_b^{\frac{b-2x}{3}} dx = \int_0^3 \left(\frac{4}{3}x^3 - 6x^2 + 12x \right) dx$$

$$= \left[\frac{x^4}{3} - \frac{6x^3}{3} + 6x^2 \right] \Big|_0^3 = 9$$

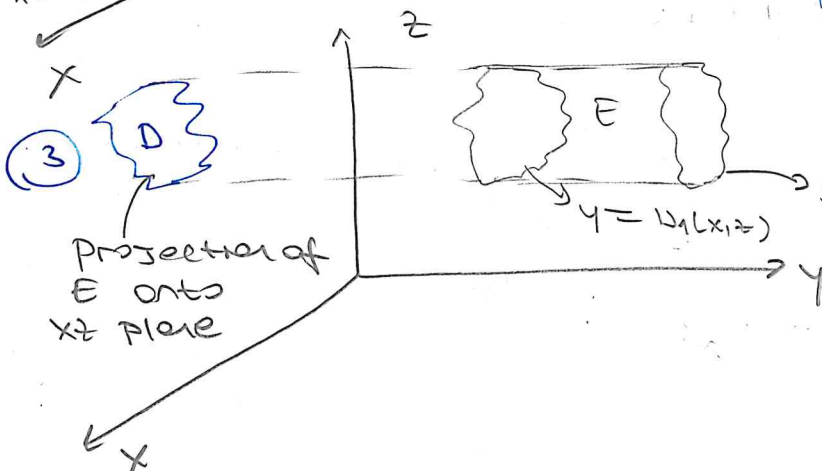
②



$$E = \{ (x,y,z) \mid (y,z) \in D, u_1(y,z) \leq x \leq u_2(y,z) \}$$

$$\iiint_E f(x,y,z) \, dV = \iint_D \left[\int_{u_1(y,z)}^{u_2(y,z)} f(x,y,z) \, dx \right] dA$$

③



$$E = \{ (x,y,z) \mid (x,z) \in D, u_1(x,z) \leq y \leq u_2(x,z) \}$$

$$\iiint_E f(x,y,z) \, dV = \iint_D \left[\int_{u_1(x,z)}^{u_2(x,z)} f(x,y,z) \, dy \right] dA$$

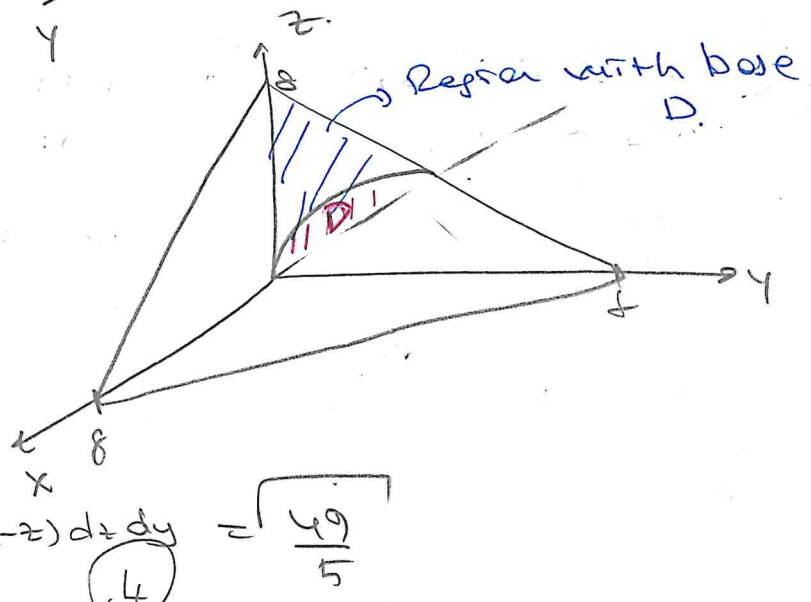
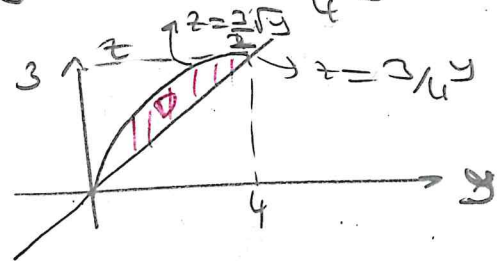
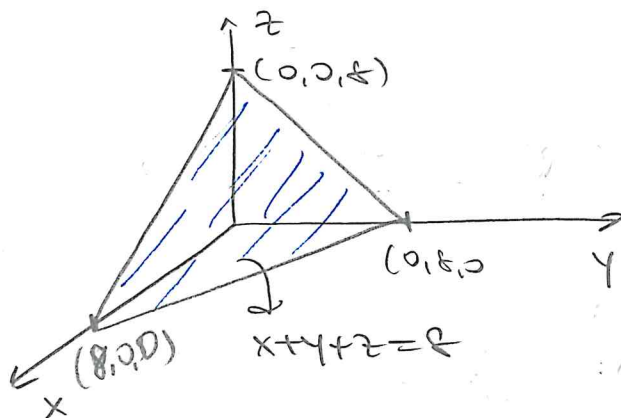
REMARK Recall that if

- $f(x) \geq 0$ $\int_a^b f(x) dx$ represents the area under the curve $y=f(x)$ from a to b .
- $f(x,y) \geq 0$ $\iint_D f(x,y) dA$ represents the volume under the surface $z=f(x,y)$ and above D .
- $f(x,y,z) \geq 0$ $\iiint_E f(x,y,z) dV$ represents the hypervolume of four dimensional object.
- $f(x,y,z) = 1$ for all points in D , the triple integral represents the volume of E

$$V(E) = \iiint_E dV$$

EXAMPLES

- (1) Determine the volume of the region under the plane $x+y+z=8$ and above the region in the yz -plane that is bounded by $z=\frac{3}{2}\sqrt{y}$ and $z=\frac{3}{4}y$



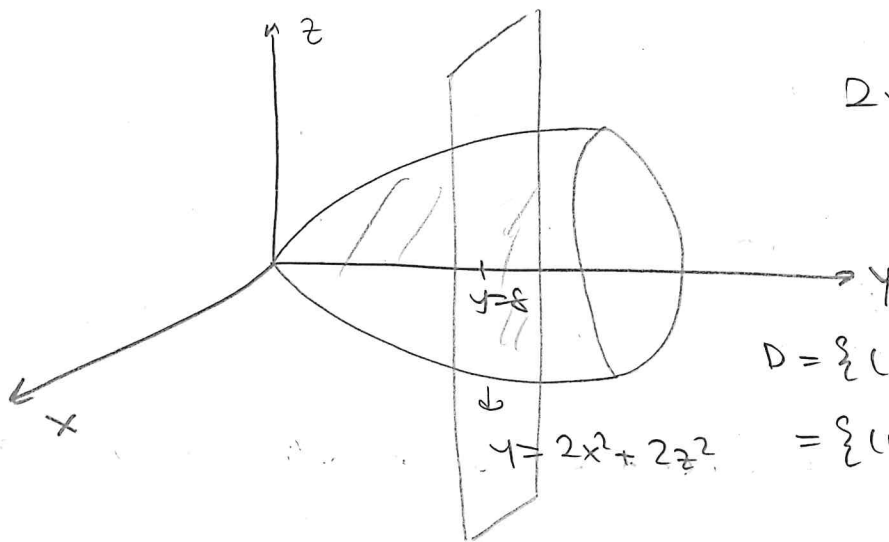
$$0 \leq y \leq 4 \quad \frac{3}{4}y \leq z \leq \frac{3}{2}\sqrt{y}$$

$$0 \leq x \leq 8 - y - z$$

$$V = \iiint_E dV = \iint_D \left[\int_0^{8-y-z} dx \right] dA$$

$$= \iint_D (8-y-z) dz dy = \int_0^4 \int_{3/4 y}^{3/2 \sqrt{y}} (8-y-z) dz dy = \frac{49}{5}$$

- ② Evaluate $\iiint_E \sqrt{3x^2 + 3z^2} \, dV$ where E is the solid bounded by $y = 2x^2 + 2z^2$ and the plane $y = 8$



$$2x^2 + 2z^2 = 8$$

$$x^2 + z^2 = 4$$

$$x = 2 \cos \theta$$

$$z = 2 \sin \theta$$

$$D = \{ (r, \theta) \mid 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi \}$$

$$= \{ (r, \theta, y) \mid 0 \leq r \leq 2, 0 \leq \theta \leq 2\pi, 2x^2 + 2z^2 \leq y \leq 8 \}$$

$$\iiint_E \sqrt{3x^2 + 3z^2} \, dV = \iint_D \left[\int_{2x^2 + 2z^2}^8 \sqrt{3x^2 + 3z^2} \, dy \right] dA$$

$$= \iint_D \sqrt{3(x^2 + z^2)} (8 - 2x^2 - 2z^2) dA$$

Convert to polar coordinates

$$= \int_0^{2\pi} \int_0^2 \sqrt{3r^2} (8 - 2r^2) r \, dr \, d\theta = \int_0^{2\pi} \int_0^2 \sqrt{3} (8r^2 - 2r^4) \, dr \, d\theta = \frac{256\sqrt{3}}{15}$$

- ③ Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$

$$8 - x^2 - y^2 = x^2 + 3y^2 \quad V = \iiint_D dz \, dy \, dx$$

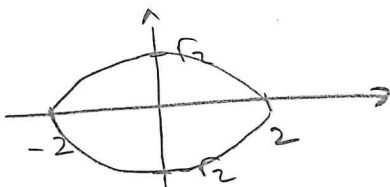
$$8 = 2x^2 + 4y^2$$

$$4 = x^2 + 2y^2$$

$$= \iiint_D (8 - x^2 - y^2 - x^2 - 3y^2) \, dz \, dy \, dx$$

$$= \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} ((8 - x^2 - y^2) - (x^2 + 3y^2)) \, dy \, dx$$

$$\int_{-2}^2 \left((8 - 2x^2)y - \frac{4}{3}y^3 \right) \Big|_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx = \int_{-2}^2 \left[8 \left(\frac{4-x^2}{2} \right)^{3/2} - \frac{8}{3} \left(\frac{4-x^2}{2} \right)^{3/2} \right] dx = 8\pi\sqrt{2}$$



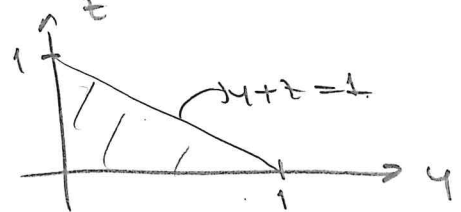
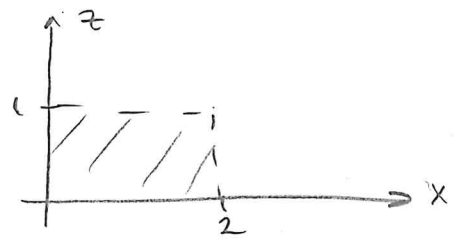
Ex Change orders of the following integrals.

(1) $\int_0^1 \int_0^{1-z} \int_0^2 dx dy dz = ?$

$0 \leq x \leq 2 \quad 0 \leq y \leq 1-z \Rightarrow 0 \leq z \leq 1$
 $0 \leq y+z \leq 1$

$\int_0^2 \int_0^1 \int_0^{1-z} dy dz dx = \int_0^1 \int_0^2 \int_0^{1-z} dy dx dz$

$= \int_0^2 \int_0^1 \int_0^{1-y} dz dy dx = \int_0^1 \int_0^2 \int_0^{1-y} dz dx dy$



(2) $\int_0^1 \int_{\sqrt{x}}^1 \int_0^{1-y} dz dy dx$

$\left. \begin{aligned} 0 \leq z \leq 1-y \\ \sqrt{x} \leq y \leq 1 \\ 0 \leq x \leq 1 \end{aligned} \right\} \text{ (A1)}$

NOTE we have $0 \leq z \leq 1-y$

Since $y \leq 1$ and $y \geq 0$
 we have $1-y \leq 1$, so, we have
 $0 \leq z \leq 1-y \leq 1$. Also, we have
 $x \geq 0 \quad \sqrt{x} \leq y \quad x \leq y^2$. Since, we
 have. $y \leq 1-z$ or $x \leq (1-z)^2$.

Now we get

$\left. \begin{aligned} 0 \leq z \leq 1 \\ 0 \leq x \leq (1-z)^2 \\ \sqrt{x} \leq y \leq 1-z \end{aligned} \right\} \text{ (A2)}$

Now to check (A1) and (A2) equal?

Consider (A2) is satisfied. Then, $0 \leq x \leq (1-z)^2 \leq 1$.

Since $0 \leq z \leq 1$, hence $0 \leq x \leq 1$. Next $\sqrt{x} \leq y \leq 1-z$.

So, $\sqrt{x} \leq y \leq 1$. Finally, $0 \leq z$ and $y \leq 1-z \quad z \leq 1-y$

This means, $0 \leq z \leq 1-y$.

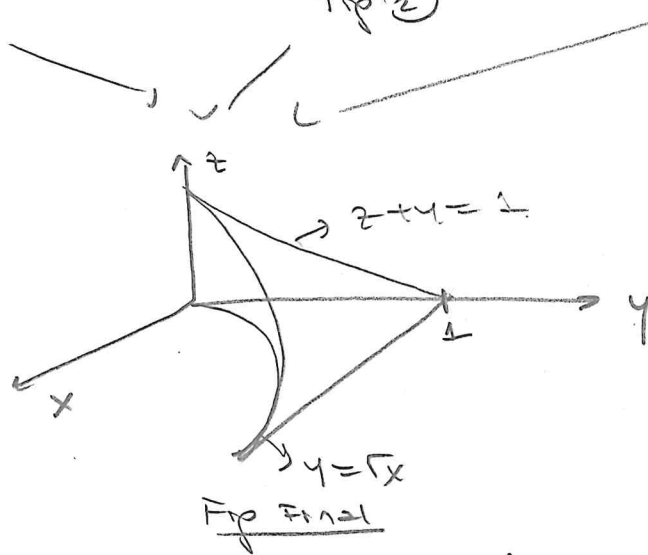
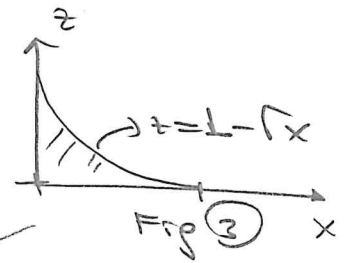
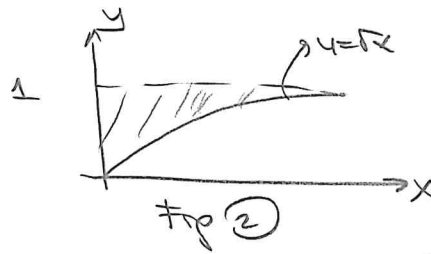
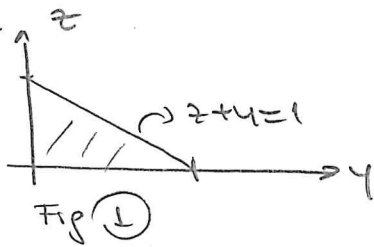


Fig 3
+
Fig Final

$$\int_0^1 \int_0^{1-\sqrt{x}} \int_{\sqrt{x}}^{1-z} dy dz dx = \int_0^1 \int_0^{(1-z)^2} \int_{\sqrt{x}}^{1-z} dy dx dz$$

Fig 1
+
Fig Final

$$\int_0^1 \int_0^{1-z} \int_0^{y^2} dx dy dz = \int_0^1 \int_0^{1-y} \int_0^{(1-z)^2} dx dz dy$$

Fig 2
+
Fig Final

$$\int_0^1 \int_0^{y^2} \int_0^{1-\sqrt{x}} dz dx dy = \int_0^1 \int_{\sqrt{x}}^1 \int_0^{1-y} dz dy dx$$

(3)

$$\int_0^1 \int_{-1}^0 \int_0^{y^2} dz dy dx$$

Fig 2

$$0 \leq z \leq y^2$$

$$-1 \leq y \leq 0$$

$$0 \leq x \leq 1$$

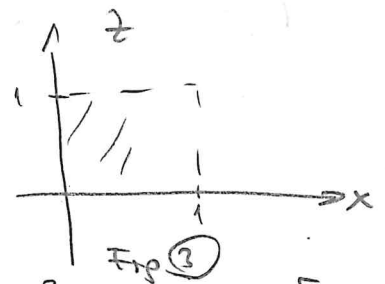
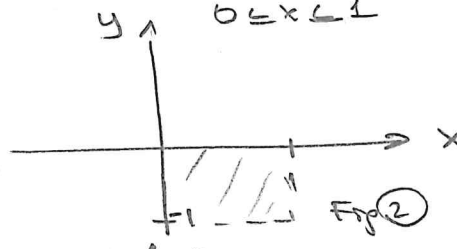
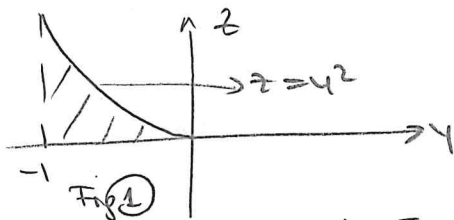


Fig 3

$$\int_0^1 \int_{-1}^0 \int_0^{y^2} dy dz dx = \int_0^1 \int_{-1}^0 \int_0^{1-\sqrt{z}} dy dz dx$$

Fig 1

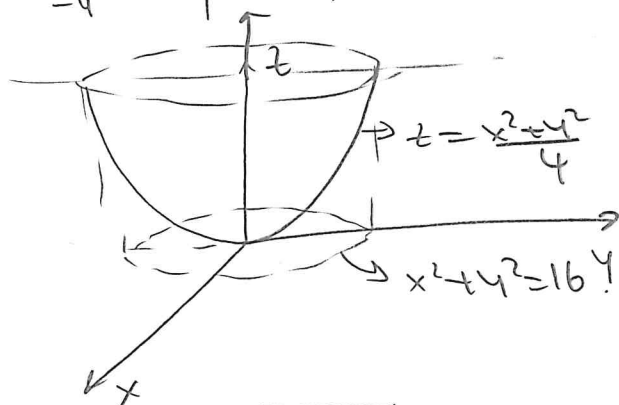
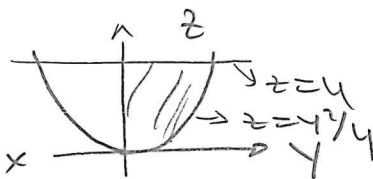
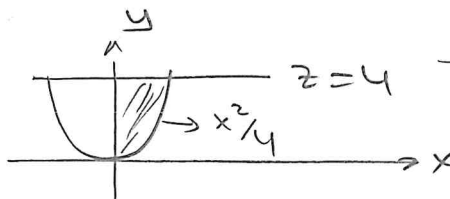
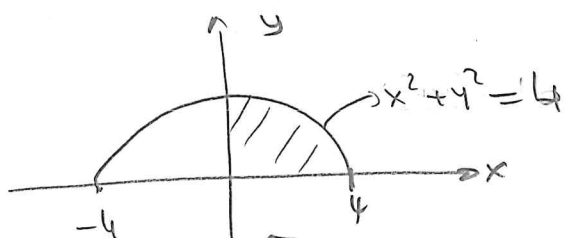
$$\int_{-1}^0 \int_0^{y^2} \int_0^1 dx dz dy = \int_{-1}^0 \int_0^{1-\sqrt{z}} \int_0^1 dx dz dy$$

④ $\int_0^4 \int_0^{\sqrt{16-x^2}} \int_{\frac{x^2+y^2}{4}}^4 dz dy dx$

$$\frac{x^2+y^2}{4} \leq z \leq 4 \Rightarrow x^2 \leq 4z - y^2$$

$$0 \leq y \leq \sqrt{16-x^2}$$

$$0 \leq x \leq 4$$



$$\int_0^4 \int_0^{\sqrt{4z}} \int_0^{\sqrt{4z-x^2}} dy dx dz = \int_0^4 \int_{y^2/4}^4 \int_0^{\sqrt{4z-y^2}} dx dz dy$$

Average Value of a Function in Space

→ The average value of a function F over a region D in space is defined by the formula

$$\text{Average value of } F \text{ over } D = \frac{1}{\text{Volume of } D} \iiint_D F dV$$

Ex 7 Find the average value of $F(x,y,z) = xyz$ throughout the cubical region D bounded by the coordinate planes and the planes $x=2$, $y=2$ and $z=2$ in the first octant

$$\int_0^2 \int_0^2 \int_0^2 xyz dx dy dz = \int_0^2 \int_0^2 \left[\frac{x^2}{2} yz \right]_{x=0}^{x=2} dy dz$$

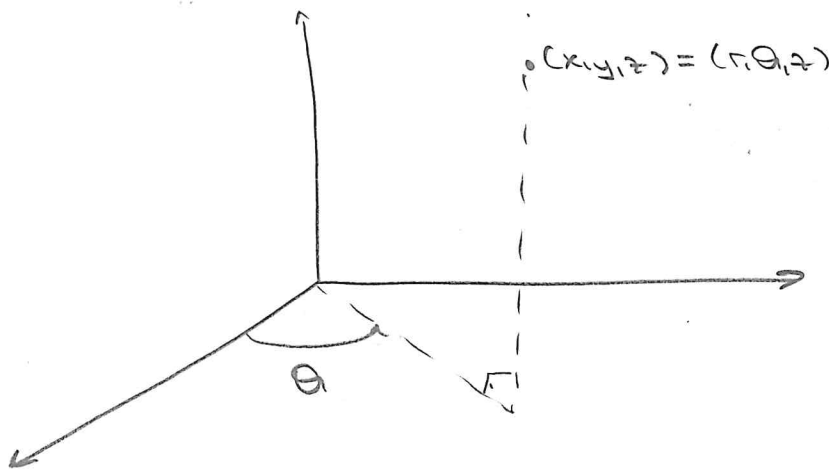
$$= \int_0^2 \int_0^2 2yz dy dz$$

$$= \int_0^2 [y^2 z]_0^2 dz = \int_0^2 4z dz = 2z^2 \Big|_0^2 = 8$$

Volume of region is $2 \cdot 2 \cdot 2 = 8$ Average = $\frac{1}{\text{Volume}} \iiint xyz dV = \frac{1}{8} \cdot 8 = 1$

CHANGE OF VARIABLES IN TRIPLE INTEGRALS

CYLINDRICAL COORDINATES



- Cylindrical coordinates are just an extension of polar coordinates into three dimensions

→ The conversion formulas for cylindrical coordinates for the region E is given.

$$E = \{ (x, y, z) \mid (x, y) \in D, u_1(x, y) \leq z \leq u_2(x, y) \} \quad (D \text{ is an } xy \text{ plane})$$

$$= \{ (r, \theta, z) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta), u_1(r \cos \theta, r \sin \theta) \leq z \leq u_2(r \cos \theta, r \sin \theta) \}$$

So, we have

$x = r \cos \theta$
 $y = r \sin \theta$
 $z = z$

$dV = r \, dz \, dr \, d\theta$

$$\iiint_E f(x, y, z) \, dV = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r \cos \theta, r \sin \theta)}^{u_2(r \cos \theta, r \sin \theta)} f(r \cos \theta, r \sin \theta, z) \, r \, dz \, dr \, d\theta$$

Ex 7 Plot the point with cylindrical coordinates $(2, 2\pi/3, 1)$ and find its rectangular coordinates

$$(r, \theta, z) = (2, 2\pi/3, 1)$$

$$x = r \cos \theta = 2 \cos(2\pi/3) = 2(-1/2) = -1$$

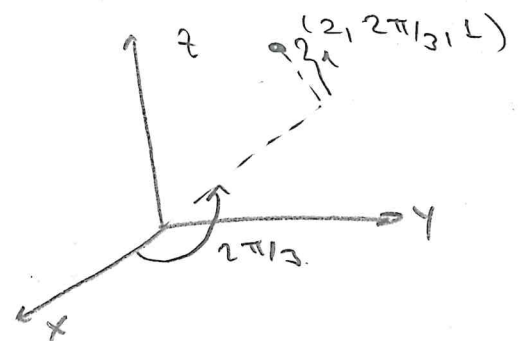
$$y = r \sin \theta = 2 \sin(2\pi/3) = 2(\sqrt{3}/2) = \sqrt{3}$$

$$z = 1$$

$$(2, 2\pi/3, 1) \longrightarrow (-1, \sqrt{3}, 1)$$

Cylindrical coordinates

Cartesian coordinates
(9)



Ex 7 Find cylindrical coordinates of the rectangular point $(3, -3, -7)$

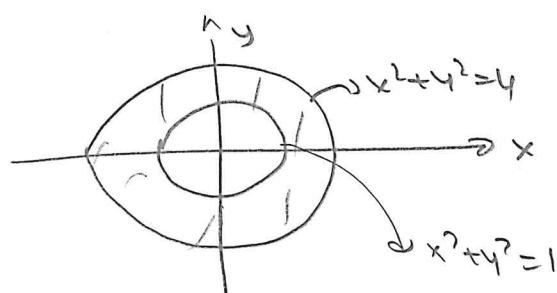
$$(x, y, z) = (3, -3, -7) \Rightarrow \begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \begin{cases} \tan \theta = -1 \\ \theta = -\pi/4 + 2k\pi \quad k \in \mathbb{Z} \end{cases}$$

$$x^2 + y^2 = r^2 \Rightarrow r^2 = 18 \quad r = 3\sqrt{2}$$

$$(x, y, z) = (3, -3, -7) \Rightarrow (r, \theta, z) = (3\sqrt{2}, -\pi/4, -7)$$

Ex 7 Evaluate $\iiint_E y \, dV$ where E is the region lies below

the plane $z = x + 2$, above the xy plane and between cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$

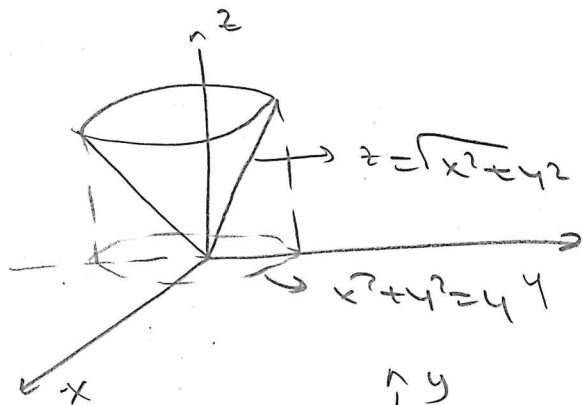


$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ z &= x + 2 = r \cos \theta + 2 \\ 0 &\leq z \leq x + 2 \end{aligned}$$

$$\int_0^{2\pi} \int_1^2 \int_0^{r \cos \theta + 2} \underbrace{r \sin \theta}_{y} r \, dz \, dr \, d\theta =$$

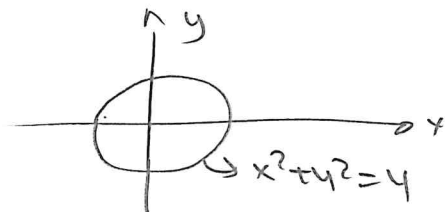
Ex 7 Evaluate $\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{\sqrt{x^2+y^2}}^2 (x^2 + y^2) \, dz \, dy \, dx$

$$E = \{ (x, y, z) \mid -2 \leq x \leq 2, -\sqrt{4-x^2} \leq y \leq \sqrt{4-x^2}, \sqrt{x^2+y^2} \leq z \leq 2 \}$$

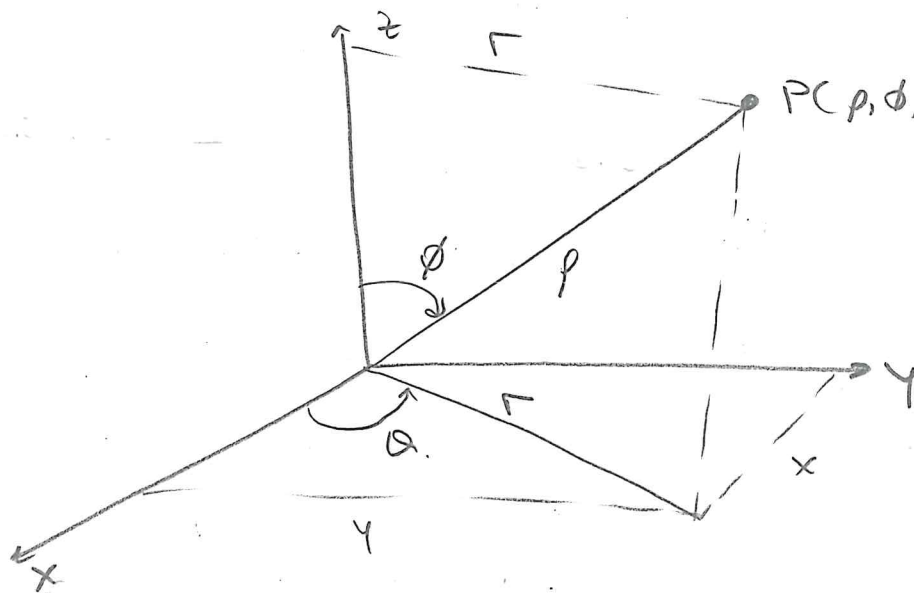


$$z = \sqrt{x^2 + y^2} \quad \wedge \quad z = 2$$

$$\int_0^{2\pi} \int_0^2 \int_r^2 r^2 r \, dz \, dr \, d\theta = \frac{16\pi}{5}$$



SPHERICAL COORDINATES



- $\rho = |\text{OP}|$ distance from origin to P

$P(\rho, \phi, \theta)$ - θ same angle as in cylindrical coordinates from positive x-axis through the positive y-axis

ϕ angle between the positive z axis and the line segment OP
 $\rho \geq 0 \quad 0 \leq \phi \leq \pi$

→ For one-to-one correspondence between rectangular coordinate system to spherical coordinate system we take

$$0 \leq \phi \leq \pi$$

$$0 \leq \theta \leq 2\pi$$

$$\frac{z}{\rho} = \cos \phi \Rightarrow z = \rho \cos \phi \quad \frac{r}{\rho} = \sin \phi \Rightarrow r = \rho \sin \phi$$

we also have $x = r \cos \theta, y = r \sin \theta$

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$

$$\rho^2 = x^2 + y^2 + z^2$$

$$(x^2 + y^2 + z^2 = \rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta + \rho^2 \cos^2 \phi = \rho^2)$$

$$E = \{(\rho, \theta, \phi) \mid a \leq \rho \leq b \quad \alpha \leq \theta \leq \beta \quad c \leq \phi \leq d\}$$

$$\iiint_E f(x, y, z) \, dV = \int_a^b \int_\alpha^\beta \int_c^d f(\rho \sin \phi \sin \theta, \rho \sin \phi \cos \theta, \rho \cos \phi) \rho^2 \sin \phi \, d\phi \, d\theta \, d\rho$$

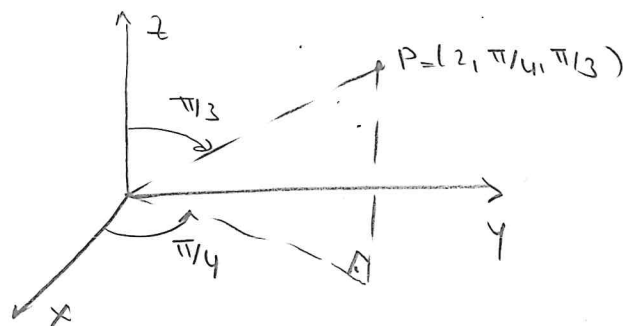
Ex 9 The point $(2, \pi/4, \pi/3)$ is given in spherical coordinates. plot the point and find its rectangular coordinates.

$$(\rho, \theta, \phi) = (2, \pi/4, \pi/3)$$

$$x = \rho \sin \phi \cos \theta = 2 \sin(\pi/3) \cos(\pi/4) \\ = 2 \cdot \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2} = \frac{\sqrt{6}}{2}$$

$$y = \rho \sin \phi \sin \theta = 2 \sin(\pi/3) \sin(\pi/4) = \sqrt{6}/2$$

$$z = \rho \cos \phi = 2 \cos(\pi/3) = 1$$



Spherical $(\rho, \theta, \phi) = (2, \pi/4, \pi/3) \longrightarrow (\frac{\sqrt{6}}{2}, \frac{\sqrt{6}}{2}, 1)$
Cartesian coordinates

(b) The point $(0, 2\sqrt{3}, -2)$ is given in the rectangular coordinates. Find spherical coordinates for this point.

$$(x, y, z) = (0, 2\sqrt{3}, -2) \Rightarrow \rho^2 = x^2 + y^2 + z^2 = 0 + 12 + 4 = 16 \quad \rho = 4$$

$$z = \rho \cos \phi \Rightarrow \cos \phi = -2/4 = -1/2 \quad \phi = 2\pi/3$$

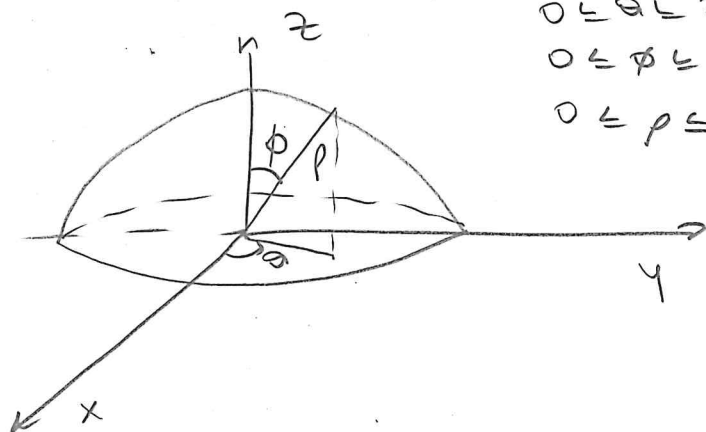
$$x = \rho \sin \phi \cos \theta \Rightarrow \cos \theta = \frac{x}{\rho \sin \phi} = 0 \Rightarrow \theta = \pi/2 \quad (\theta \neq 3\pi/2 \text{ since } y = r \sin \theta = 2\sqrt{3})$$

for 1-1 correspondence we will not deal with $\theta = \pi/2 + 2\pi$.

$$(\rho, \theta, \phi) = (4, \pi/2, 2\pi/3)$$

Ex 7 Evaluate $\iiint_E 16z \, dV$ where E is the upper half of

the sphere $x^2 + y^2 + z^2 = 1$
 $0 \leq \theta \leq 2\pi$
 $0 \leq \phi \leq \pi/2$
 $0 \leq \rho \leq 1$



$$\iiint_E 16z \, dV = \int_0^{\pi/2} \int_0^{2\pi} \int_0^1 16 \rho \cos \phi \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi$$

$$= \int_0^{\pi/2} \int_0^{2\pi} \int_0^1 16 \rho^3 \sin 2\phi \, d\rho \, d\theta \, d\phi$$

$$= \int_0^{\pi/2} \int_0^{2\pi} 2 \sin 2\phi \, d\theta \, d\phi$$

$$= \int_0^{\pi/2} 4\pi \sin 2\phi \, d\phi$$

$$= -2\pi \cos 2\phi \Big|_0^{\pi/2} =$$

$$= -2\pi(-1 - 1) = -4\pi$$

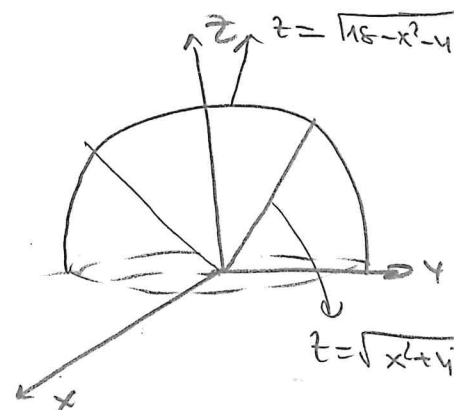
(12)

Ex 7 Convert

$$\int_0^3 \int_0^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{16-x^2-y^2}} (x^2+y^2+z^2) dz dx dy \quad \text{Convert to spherical coordinates}$$

$$E = \{(x, y, z) \mid 0 \leq y \leq 3 \quad 0 \leq x \leq \sqrt{9-y^2} \quad \sqrt{x^2+y^2} \leq z \leq \sqrt{16-x^2-y^2}\}$$

We have a portion of the right half of a disk radius 3 centered at origin.



$$z = \sqrt{16-x^2-y^2} \Rightarrow x^2+y^2+z^2 = 16 = \rho^2$$

$$0 \leq \rho \leq \sqrt{16} = 4$$

The lower bound $z = \sqrt{x^2+y^2}$ is the upper half of the cone.

To find the range for ϕ , we will use the intersection of the cone and sphere.

$$z = \sqrt{x^2+y^2}$$

$$x^2+y^2+z^2 = 16 \Rightarrow (\sqrt{x^2+y^2})^2 + z^2 = 16 \quad 2z^2 = 16$$

$$z = 2$$

(z is positive since we have the upper half of the cone and sphere)

$$z = \rho \cos \phi = 2$$

$$4 \cos \phi = 2 \Rightarrow \cos \phi = 1/2$$

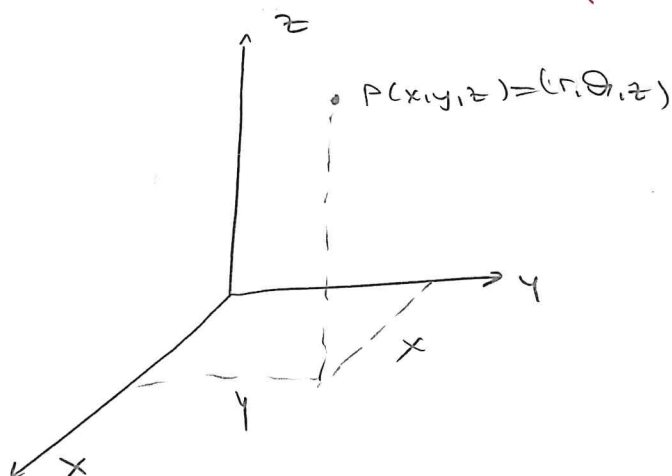
$$\phi = \pi/4$$

$$0 \leq \phi \leq \pi/4$$

$$\int_0^3 \int_0^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{16-x^2-y^2}} (x^2+y^2+z^2) dz dx dy = \int_0^{\pi/4} \int_0^{\pi/2} \int_0^{4} \rho^2 \sin \phi d\rho d\phi dy$$

SUMMARY PAGE

CYLINDRICAL COORDINATES



$$x^2 + y^2 = r^2$$

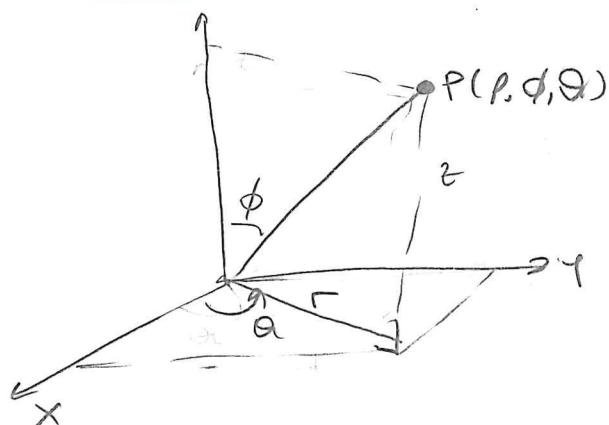
$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$dV = r \, dz \, dr \, d\theta$$

SPHERICAL COORDINATES



$$x^2 + y^2 + z^2 = \rho^2$$

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

Cylindrical

$$\iiint_E F(x, y, z) \, dV = \int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} \int_{h_1(\theta)}^{h_2(\theta)} F(r \cos \theta, r \sin \theta, z) \, r \, dz \, dr \, d\theta$$

Spherical

$$\iiint_E F(x, y, z) \, dV = \int_{\theta_1}^{\theta_2} \int_{\phi_1(\theta)}^{\phi_2(\theta)} \int_{\rho_1(\theta, \phi)}^{\rho_2(\theta, \phi)} F(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \, \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

CHANGE COORDINATES

Cylindrical to Cartesian

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

Spherical to Cartesian

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

Spherical to Cylindrical

$$r = \rho \sin \phi$$

$$z = \rho \cos \phi$$

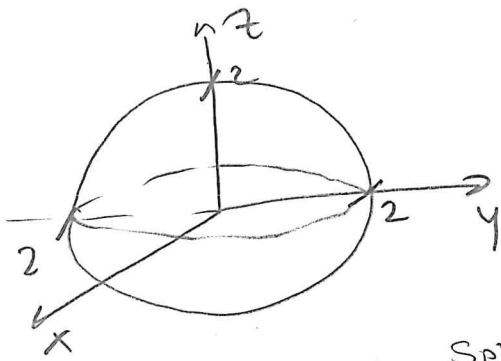
$$\theta = \theta$$

$$dV = dx \, dy \, dz \text{ (Cartesian)}$$

$$dV = r \, dz \, dr \, d\theta \text{ (Cylindrical)}$$

$$dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \text{ (Spherical)}$$

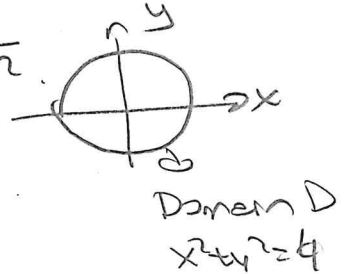
Ex Set up triple integrals for the volume of the sphere $\rho=2$ (a) Spherical (b) Cylindrical (c) Rectangular



(a) $V = \int_0^{2\pi} \int_0^\pi \int_0^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$

(b) $V = 2 \int_0^{2\pi} \int_0^2 \int_0^{\sqrt{4-r^2}} r \, dz \, dr \, d\theta$

Sphere $x^2 + y^2 + z^2 = 4$ $z = \sqrt{4-r^2}$



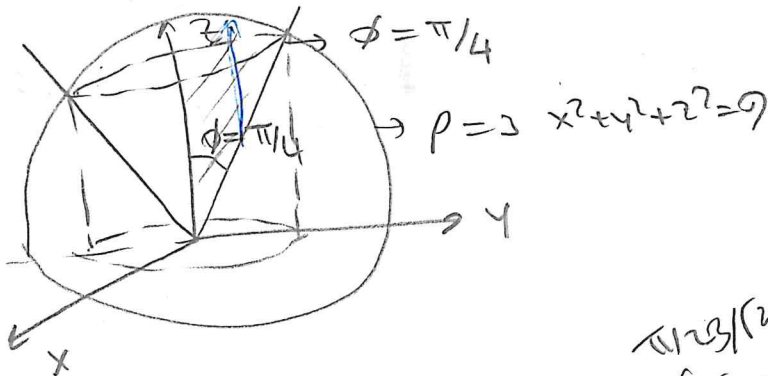
(c) $V = 2 \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{\sqrt{4-x^2-y^2}} dz \, dy \, dx$

Ex Let D be the region in the first octant that is bounded below the cone $\phi = \pi/4$ and above the sphere $\rho=3$ express the volume

(a) Cylindrical

(b) Spherical

(c) Compute one of them.



(a) Cylindrical Parallels to z

Equation of cone $\phi = \pi/4$

$$\left. \begin{aligned} z &= \rho \sin \phi \\ r &= \rho \cos \phi \end{aligned} \right\} z = r \text{ for } \pi/4$$

Equation of sphere

$x^2 + y^2 + z^2 = 9$

$z^2 = 9 - x^2 - y^2$

$z = \sqrt{9 - x^2 - y^2}$

$z = \sqrt{9 - r^2}$

So $r \leq z \leq \sqrt{9-r^2}$

(16)

$\int_0^{2\pi} \int_0^{\pi/2} \int_0^{\sqrt{9-r^2}} r \, dz \, dr \, d\theta$

$r = \rho \sin \phi$ $\phi = \pi/4$
 $\rho = 3$

$r = 3/\sqrt{2}$
 $0 \leq r \leq 3/\sqrt{2}$

In the first octant
 $x, y, z \geq 0$

So
 $0 \leq \theta \leq \pi/2$



$0 \leq \theta \leq \pi/2$

(b) Spherical

$$\int_0^{\pi/2} \int_0^{\pi/4} \int_0^3 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$(c) \quad V = \int_0^{\pi/2} \int_0^{\pi/4} \int_0^3 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = 9 \int_0^{\pi/2} \int_0^{\pi/4} \sin \phi \, d\phi \, d\theta$$

$$= 9 \cdot \int_0^{\pi/2} (-\cos \phi) \Big|_0^{\pi/4} d\theta = 9 \int_0^{\pi/2} (-\cos(\pi/4) + \cos 0) d\theta$$

$$= 9\pi \left(\frac{2 - \sqrt{2}}{4} \right)$$

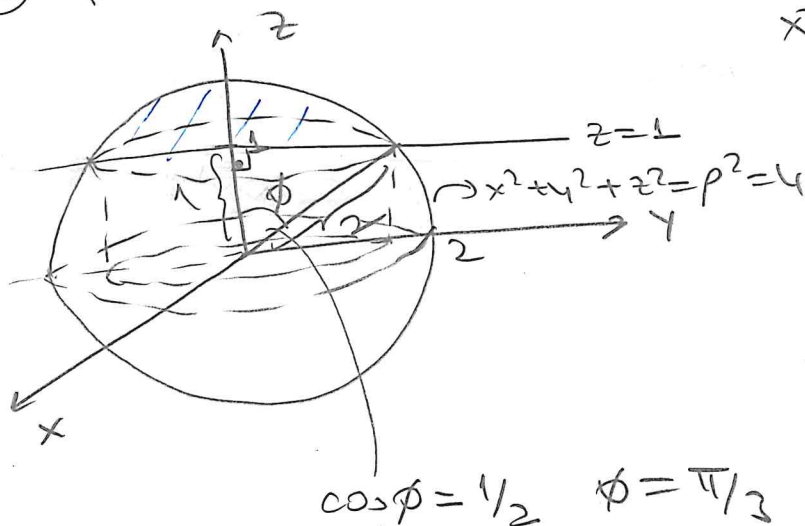
Ex 7 Let D be the smaller ^{cap} cut from a solid ball of radius 2 units by a plane 1 unit from the center of the sphere express the volume of D integral.

(a) Spherical

(b) Cylindrical

(c) rectangular

(d) find volume



$$x^2 + y^2 + z^2 = 4 \wedge z \geq 1$$

$$x^2 + y^2 = 3 \quad \sqrt{0 \leq r \leq \sqrt{3}}$$

$$0 \leq \theta \leq 2\pi$$

$$z = \rho \cos \phi \wedge z = 1$$

$$\rho = 1/\cos \phi$$

$$\sqrt{0 \leq \rho \leq 1/\cos \phi}$$

$$\sqrt{0 \leq \phi \leq \pi/3}$$

$$\cos \phi = 1/2 \quad \phi = \pi/3$$

(a) Spherical

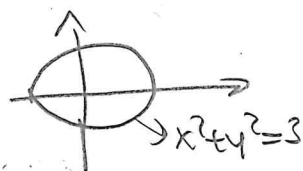
$$\int_0^{2\pi} \int_0^{\pi/3} \int_{\rho=1/\cos \phi}^2 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

(b) Cylindrical

$$\int_0^{2\pi} \int_{z=1}^{\sqrt{3}} \int_{r=0}^{\sqrt{4-r^2}} r \, dz \, dr \, d\theta$$

(c) Rectangular

$$\int_{-3}^3 \int_{-\sqrt{3-y^2}}^{\sqrt{3-y^2}} \int_{z=1}^{\sqrt{4-x^2-y^2}} dz \, dx \, dy$$



(1b)

② Volume from cylindrical

$$\int_0^{2\pi} \int_0^{\sqrt{3}} \int_{z=1}^{\sqrt{4-r^2}} r \, dz \, r \, d\theta = \int_0^{2\pi} \int_0^{\sqrt{3}} r (\sqrt{4-r^2} - 1) \, dr \, d\theta$$

$$\int \sqrt{4-r^2} \cdot r \, dr = -\frac{1}{2} \frac{u^{3/2}}{3/2} = -\frac{1}{3} (4-r^2)^{3/2}$$

$$4-r^2 = u$$

$$-2r \, dr = du$$

$$= \int_0^{2\pi} \left(-\frac{1}{3} (4-r^2)^{3/2} - \frac{r^2}{2} \right) \Big|_0^{\sqrt{3}} d\theta$$

$$= \int_0^{2\pi} \left(\left(-\frac{1}{3} (4-3)^{3/2} - \frac{3}{2} \right) - \left(-\frac{4^{3/2}}{3} \right) \right) d\theta$$

$$= \left(\left(-\frac{1}{3} - \frac{3}{2} \right) + \frac{4^{3/2}}{3} \right) \cdot 2\pi = \left(-\frac{1}{3} - \frac{3}{2} \right) + \frac{2^{2 \cdot 3/2}}{3} \cdot 2\pi$$

$$= \left(-\frac{7}{6} - \frac{3}{2} \right) \cdot 2\pi = \left(-\frac{14-9}{6} \right) \cdot 2\pi$$

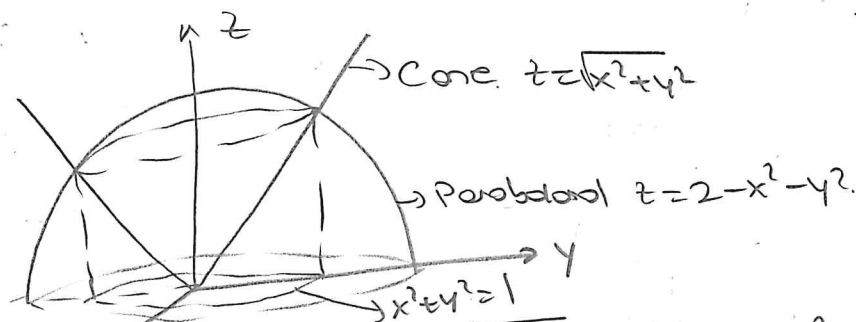
$$= \frac{5}{6} \cdot 2\pi$$

Ex 7 Let D be the region bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the paraboloid $z = 2 - x^2 - y^2$. Evaluate the following integral

$$\iiint_D \sqrt{x^2 + y^2} \, dV$$

$$z = \rho \cos \phi$$

$$z = \sqrt{x^2 + y^2}$$



Boundaries $\sqrt{x^2 + y^2} \leq z \leq 2 - x^2 - y^2$

$$r \leq z \leq 2 - r^2$$

$$0 \leq r \leq 1$$

$$0 \leq \theta \leq 2\pi$$

$$x^2 + y^2 = \rho^2 \sin^2 \phi$$

$$z = 2 - x^2 - y^2 \wedge z = \sqrt{x^2 + y^2}$$

$$z = 2 - r^2 \wedge z = r$$

$$2 - r^2 = r$$

$$r^2 + r - 2 = 0$$

$$(r+2)(r-1) = 0$$

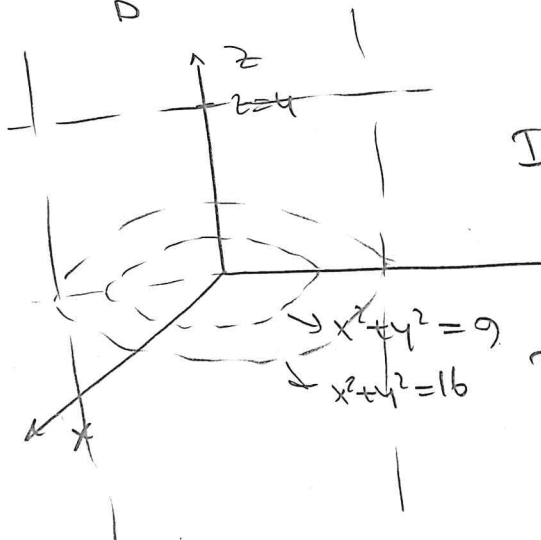
$$r = -2 \wedge r = 1$$

(17)

$$\begin{aligned}
 \iiint_D \sqrt{x^2+y^2} \, dV &= \int_0^{2\pi} \int_0^1 \int_r^{2-r^2} r \cdot r \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^1 r^2 z \Big|_r^{2-r^2} \, dr \, d\theta \\
 &= \int_0^{2\pi} \int_0^1 r^2 (2-r^2-r) \, dr \, d\theta = \int_0^{2\pi} \left(\frac{2r^3}{3} - \frac{r^5}{5} - \frac{r^4}{4} \right) \Big|_0^1 \, d\theta \\
 &= \int_0^{2\pi} \left(\frac{2}{3} - \frac{1}{5} - \frac{1}{4} \right) \, d\theta = \frac{26\pi}{60}
 \end{aligned}$$

Ex Let D be the region bounded by $x^2+y^2=9$, $x^2+y^2=16$, cylinders and $z=0$ and $z=4$ planes then evaluate

$\iiint_D x^2 \, dV$



$x^2+y^2=16 \Rightarrow r=4$
 $x^2+y^2=9 \Rightarrow r=3$

$$I = \int_0^{2\pi} \int_{r=3}^{r=4} \int_{z=0}^{z=4} r \cdot r^2 \cos^2 \theta \, dz \, dr \, d\theta$$

$$I = \int_0^{2\pi} \int_{r=3}^{r=4} r^3 \cos^2 \theta \cdot (4-0) \, dr \, d\theta$$

$$= \int_0^{2\pi} \frac{r^4}{4} \cos^2 \theta \Big|_{r=3}^{r=4} \, d\theta$$

$$= 175 \int_0^{2\pi} \frac{(1+\cos 2\theta)}{2} \, d\theta$$

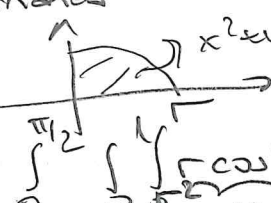
$$= 175 \left(\frac{\theta}{2} + \frac{\sin 2\theta}{2} \right) \Big|_0^{2\pi}$$

Ex $\int_0^1 \int_0^{\sqrt{1-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{x^2+y^2}} x y z \, dz \, dx \, dy$

write in cylindrical coordinates

$x^2+y^2 \leq z \leq \sqrt{x^2+y^2}$
 $0 \leq x \leq \sqrt{1-y^2}$
 $0 \leq y \leq 1$

$r^2 \leq z \leq r$
 $x^2+y^2=1$
 $0 \leq r \leq 1$
 $0 \leq \theta \leq 2\pi$

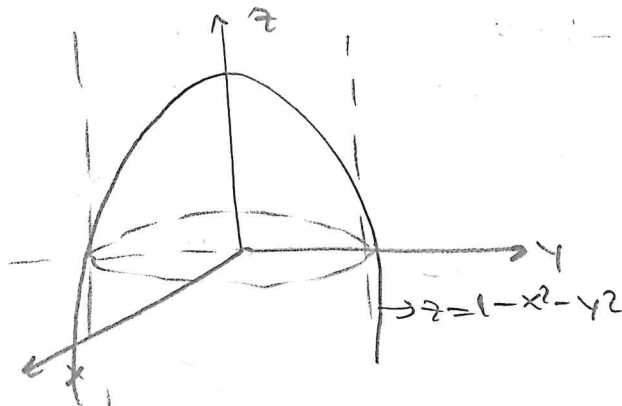


$\int_0^{\pi/2} \int_0^1 \int_{r^2}^r r \cos \theta \cdot r \sin \theta \cdot (z) \cdot r \, dz \, dr \, d\theta$

Don't forget to keep z .

ex Let D be the region above $z=0$ plane, below $z=1-x^2-y^2$ paraboloid and inside the cylinder $x^2+y^2=1$. Then, evaluate the following integral

$$\iiint_D \sqrt{x^2+y^2} \, dV$$



$$z=0 \text{ and } z=1-r^2$$

$$\int_0^{2\pi} \int_0^1 \int_0^{1-r^2} r \, dz \, dr \, d\theta$$

$$z=1-x^2-y^2 \wedge z \geq 0$$

$$1-x^2-y^2=0$$

$$x^2+y^2=1$$

$$0 \leq r \leq 1 \quad 0 \leq \theta \leq 2\pi$$

$$= \int_0^{2\pi} \int_0^1 \int_0^{1-r^2} r^2 \, dz \, dr \, d\theta = \int_0^{2\pi} \int_0^1 r^2 \cdot z \Big|_0^{1-r^2} \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^1 r^2 (1-r^2) \, dr \, d\theta = \int_0^{2\pi} \left(\frac{r^3}{3} - \frac{r^5}{5} \right) \Big|_0^1 \, d\theta = 4\pi/15$$

Cylindrical Coordinates

$$\int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_0^{1-x^2-y^2} \sqrt{x^2+y^2} \, dz \, dy \, dx$$

Spherical

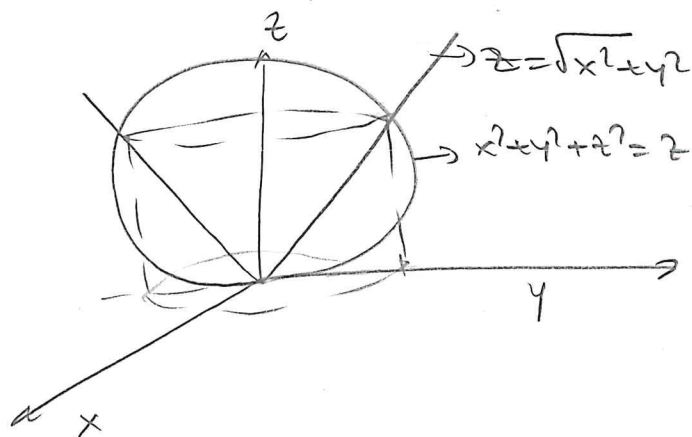
$$0 \leq \phi \leq \pi/2$$

$$0 \leq \theta \leq 2\pi$$

$$z = \rho \cos \phi$$

$$\rho \cos \phi = 1 - \rho^2 \sin^2 \phi \quad \text{Difficult to simplify}$$

Ex 7 Use spherical coordinates to find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 2$.



$$x^2 + y^2 + z^2 = \rho^2 \quad z = \rho \cos \phi$$

$$x^2 + y^2 + z^2 = 2$$

$$\rho = \sqrt{2}$$

$$z = \sqrt{x^2 + y^2}$$

$$\rho \cos \phi = \rho \sin \phi$$

$$\phi = \pi/4$$

$$\int_0^{2\pi} \int_0^{\pi/4} \int_0^{\sqrt{2}} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi/4} \sin \phi \left[\frac{\rho^3}{3} \right]_0^{\sqrt{2}} d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/4} \sin \phi \frac{\cos^3 \phi}{3} d\phi \, d\theta = \left[-\frac{\cos^4 \phi}{12} \right]_0^{\pi/4} \cdot 2\pi = \pi/6$$

Ex 8 Let $\Omega = \{(x, y, z) : a^2 \leq x^2 + y^2 + z^2 \leq b^2\}$ then evaluate.

$$I = \iiint_{\Omega} \frac{x^2 + y^2}{x^2 + y^2 + z^2} \, dx \, dy \, dz$$

Spherical coordinates

$$x = \rho \sin \phi \cos \theta$$

$$y = \rho \sin \phi \sin \theta$$

$$z = \rho \cos \phi$$

$$x^2 + y^2 + z^2 = \rho^2$$

$$x^2 + y^2 = \rho^2 \sin^2 \phi$$

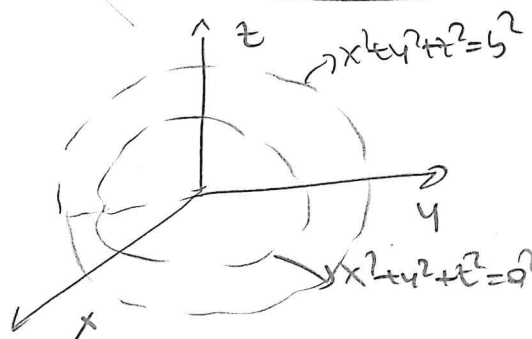
$$\int_0^{2\pi} \int_0^{\pi} \int_a^b \frac{\rho^2 \sin^2 \phi}{\rho^2} \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \sin^3 \phi \left[\frac{\rho^3}{3} \right]_a^b d\phi \, d\theta$$

$$= \frac{b^3 - a^3}{3} \int_0^{2\pi} \int_0^{\pi} (1 - \cos^2 \phi) \sin \phi \, d\phi \, d\theta$$

$$= \frac{b^3 - a^3}{3} \int_0^{2\pi} \left(-\cos \phi + \frac{1}{3} \cos^3 \phi \right) \Big|_0^{\pi} d\theta$$

$$= \frac{b^3 - a^3}{3} \cdot \frac{4}{3} \int_0^{2\pi} d\theta = \frac{8\pi}{9} (b^3 - a^3)$$



Ex Let Ω be the region between $x^2+y^2+z^2-2az=0$ and $x^2+y^2+z^2-at=0$ spheres. Then, evaluate

$$I = \iiint \tau \, dx \, dy \, dz =$$

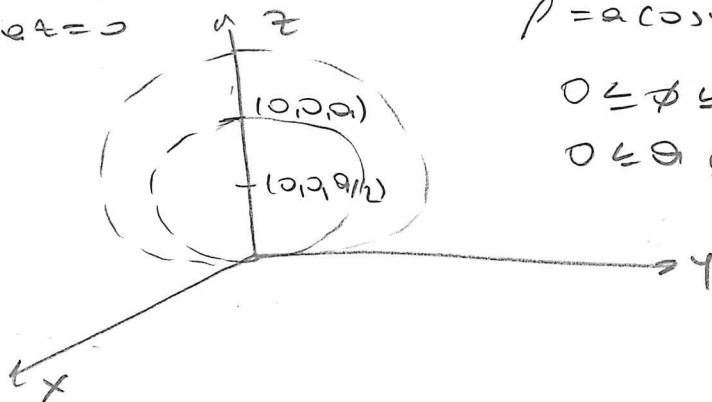
$$x^2+y^2+z^2-2az=0 \quad \rho^2-2a\rho\cos\phi=0 \quad \rho=2a\cos\phi$$

$$x^2+y^2+z^2-at=0$$

$$\rho=a\cos\phi$$

$$0 \leq \phi \leq \pi/2$$

$$0 \leq \theta \leq 2\pi$$



$$I = \int_0^{2\pi} \int_0^{\pi/2} \int_{a\cos\phi}^{2a\cos\phi} \underbrace{\rho \cos\phi}_{\tau} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi/2} \frac{\rho^4}{4} \cos\phi \sin\phi \bigg|_{a\cos\phi}^{2a\cos\phi} d\phi \, d\theta$$

$$= \frac{1}{4} \int_0^{2\pi} \int_0^{\pi/2} (16a^4\cos^5\phi - a^4\cos^5\phi) \sin\phi \, d\phi \, d\theta$$

$$= \frac{15a^4}{24} \int_0^{2\pi} -\cos^6\phi \bigg|_0^{\pi/2} d\theta = \frac{15a^4}{24} \int_0^{2\pi} d\theta = \frac{13}{4} a^4 \pi$$

Ex Evaluate $\iiint_B e^{\frac{(x^2+y^2+z^2)^{3/2}}{3}} \, dV$ where $B = \{(x,y,z) | x^2+y^2+z^2 \leq 1\}$

$$\int_0^{2\pi} \int_0^{\pi} \int_0^1 \frac{(\rho^2)^{3/2}}{3} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\pi} \frac{\rho^3}{3} \sin\phi \, d\phi \, d\theta$$

$$= \frac{1}{3} \int_0^{2\pi} \int_0^{\pi} (e^{\rho^3} - 1) \sin\phi \, d\phi \, d\theta = -\frac{1}{3} (e^{\rho^3} - 1) \int_0^{2\pi} \cos\phi \bigg|_0^{\pi} d\theta$$

$$= \frac{4}{3} (e^{\rho^3} - 1) \pi \quad \text{Ans.}$$

CHANGE OF VARIABLES FOR TRIPLE INTEGRALS

Consider the region R in x, y, z domain use transformation
 $x = g(u, v, w)$, $y = h(u, v, w)$ and $z = k(u, v, w)$. Here, we will
 again use Jacobian

$$J = \frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix}$$

So triple integral becomes

$$\iiint_R f(x, y, z) \, dV = \iiint_S f(g(u, v, w), h(u, v, w), k(u, v, w)) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| \, du \, dv \, dw$$

$\frac{\partial(x, y, z)}{\partial(u, v, w)} \neq 0$

Ex Verify that $dv = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$ when using spherical
 coordinate
 $x = \rho \sin \phi \cos \theta$ $y = \rho \sin \phi \sin \theta$ $z = \rho \cos \phi$

$$J = \frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \begin{vmatrix} x_\rho & x_\phi & x_\theta \\ y_\rho & y_\phi & y_\theta \\ z_\rho & z_\phi & z_\theta \end{vmatrix} = \begin{vmatrix} \sin \phi \cos \theta & \cos \phi \cos \theta & -\rho \sin \phi \sin \theta \\ \sin \phi \sin \theta & \cos \phi \sin \theta & \rho \sin \phi \cos \theta \\ \cos \phi & -\sin \phi & 0 \end{vmatrix}$$

$$= \cos \phi \left[-\rho^2 \sin \phi \cos \phi \sin^2 \theta - \rho^2 \sin \phi \cos \phi \cos^2 \theta \right]$$

$$= -\rho \sin \phi \left[\rho \sin^2 \phi \cos^2 \theta + \rho \sin^2 \phi \sin^2 \theta \right]$$

$$= -\rho^2 \sin \phi \cos^2 \phi - \rho^2 \sin \phi \sin^2 \phi = -\rho^2 \sin \phi (\cos^2 \phi + \sin^2 \phi) = -\rho^2 \sin \phi$$

So, we have $dV = |-\rho^2 \sin \phi| \, d\rho \, d\phi \, d\theta$
 $= \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$

Remember we take $0 \leq \phi \leq \pi$ since we have
 $\sin \phi \geq 0$ so $|\sin \phi| = \sin \phi$
 we don't need absolute
 value for $\sin \phi$

Ex 7

Evaluate

$$\int_0^3 \int_0^4 \int_{x=y/2}^{x=y/2+1} \left(\frac{2x-y}{2} + \frac{z}{3} \right) dx dy dz$$

apply $u = \frac{2x-y}{2}, v = y/2, w = z/3$

So, we have $x = u+v, y = 2v, z = 3w$

Boundaries

$x = y/2$	$\Rightarrow u+v = \frac{2v}{2} = v$	$\left[\begin{array}{l} u=0 \\ v=1 \\ v=0 \\ v=2 \\ v=0 \\ w=1 \end{array} \right]$
$x = y/2 + 1$	$\Rightarrow u+v = \frac{2v}{2} + 1 = v+1$	
$y = 0$	$\Rightarrow 2v = 0$	
$y = 4$	$\Rightarrow 2v = 4$	
$z = 0$	$\Rightarrow 3w = 0$	
$z = 3$	$\Rightarrow 3w = 3$	

$$J = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix} = \begin{vmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{vmatrix} = 6$$

$$\int_0^3 \int_0^4 \int_{x=y/2}^{x=y/2+1} \left(\frac{2x-y}{2} + \frac{z}{3} \right) dx dy dz = \int_{w=0}^1 \int_{v=0}^2 \int_{u=0}^1 (u+w) 6 du dv dw$$

$$= \int_0^1 \int_0^2 \int_0^1 (u+w) 6 du dv dw = 6 \int_0^1 \int_0^2 \left(\frac{u^2}{2} + wu \right) \Big|_0^1 dv dw$$

$$= 6 \int_0^1 \int_0^2 \left(\frac{1}{2} + w \right) dv dw = 6 \int_0^1 \left[\frac{v}{2} + wv \right]_0^2 dw$$

$$= 6 \int_0^1 (1+2w) dw$$

$$= 6 \left[w + w^2 \right]_0^1 = 6 \cdot 2 = 12$$

